

Grade 7  
Unit 8  
Lesson 1 Practice

Name Answer Key  
Date \_\_\_\_\_  
Period \_\_\_\_\_

1. Write the product as a repeated multiplication expression, then write the expression with an exponent.

| Product         | Repeated Multiplication                                                                                     | Power of the Form $a^b$ |
|-----------------|-------------------------------------------------------------------------------------------------------------|-------------------------|
| $2^2 \cdot 2^3$ | $(2 \cdot 2) \cdot (2 \cdot 2 \cdot 2) = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2$                                 | $2^5$                   |
| $4^2 \cdot 4^5$ | $(4 \cdot 4) \cdot (4 \cdot 4 \cdot 4 \cdot 4 \cdot 4) = 4 \cdot 4 \cdot 4 \cdot 4 \cdot 4 \cdot 4 \cdot 4$ | $4^7$                   |

2. A base is the number used as a factor in exponential form. An exponent is the number of times the base occurs as a factor.

3. Evaluate each power then multiply.

$$5^3 \cdot 5^2$$
$$125 \cdot 25 = 3,125$$

4. Apply the product of powers exponent law, then evaluate the power.

$$(-6)^2 \cdot (-6)^7$$
$$(-6)^9$$
$$-10,077,696$$

5. Leeland says that he can evaluate  $7^3 \cdot 3^7$  using the product of powers exponent law. Explain to Leeland if he is correct.

He is not because he does not have like bases.

6. Write each power of a product as a repeated multiplication expression and group repeated factors by using the commutative and associative properties. Then, write each expression with an exponent in the form  $a^m \cdot b^n$ .

| Power of a Product | Repeated Multiplication             | Group Repeated Factors                   | Power of the Form $a^m \cdot b^n$ |
|--------------------|-------------------------------------|------------------------------------------|-----------------------------------|
| $(2 \cdot 3)^3$    | $(2 \cdot 3)(2 \cdot 3)(2 \cdot 3)$ | $(2 \cdot 2 \cdot 2)(3 \cdot 3 \cdot 3)$ | $2^3 \cdot 3^3$                   |
| $(7 \cdot 8)^2$    | $(7 \cdot 8)(7 \cdot 8)$            | $(7 \cdot 7)(8 \cdot 8)$                 | $7^2 \cdot 8^2$                   |

7. Evaluate inside the parenthesis, then apply the exponent.

$$(-3 \cdot 6)^4$$

$$(-18)^4$$

$$104,796$$

8. Apply the power of a product exponent law, then evaluate the power.

$$(2 \cdot -9)^3$$

$$2^3 \cdot (-9)^3$$

$$8 \cdot (-729)$$

$$-5,832$$

9. Use the power of a power exponent law to simplify the expression.

$$[(5)^4]^4 = \underline{5^{16}}$$

10. Rewrite the expression by using the laws of exponents. Circle the exponent law used from the list provided. Then, evaluate the exponential expression.

| Expression  | Exponent Law                                                                                                                                                            | Evaluated Expression |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| $[(2)^5]^2$ | <ul style="list-style-type: none"> <li>Product of powers law</li> <li>Power of product law</li> <li><b>Power of power law</b></li> <li>Identity exponent law</li> </ul> | $2^{10}$             |